Existence of Sparsest Representations

Theorems, proofs, and the limits of what existence establishes

LRAC research project

29 October 2026

Abstract

We give a precise answer to whether every sequence has a sparsest possible representation. First, any declared representation family with at least one feasible finite code attains a minimum nonzero count, by well-ordering; compactness is unnecessary for this conditional existence result. Second, every nonzero sequence admits an exact one-coefficient representation in a sequence-dependent algebraic basis. In finite dimensions this basis can be orthonormal and constructed by Gram–Schmidt; the same conclusion holds for finite-energy infinite sequences in a sequence-dependent Hilbert basis. These are existence statements, not evidence of compression: the chosen basis can contain the sequence itself. We then prove existence in fixed dictionaries, a conditional uniqueness result, and limitations for shared stable decoders and finite-precision codes. No theorem here asserts that the trained autoencoder attains a global optimum, or that one universal, useful sparsifying domain has been identified.

1 The question and its quantifiers

A finite real recording is a vector $x \in \mathbb{R}^N$. More generally, a sequence is an element of $\mathbb{F}^I$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and I is a finite index set or $\mathbb{N}$. An infinite sequence need not belong to ℓ^2; the space and reconstruction criterion must therefore be stated explicitly. We use $\mathbb{N} = \{1, 2, \ldots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$; finite signal and dictionary dimensions are positive integers.

The word *sparsest* refers here to the **minimum number of nonzero scalar coefficients**, not minimum bit length, smallest coefficient magnitudes, or most interpretable features. We distinguish:

1. existence of a minimum in a declared feasible family;
2. existence when a new basis may be chosen for each individual sequence;
3. existence under a shared decoder, restricted adaptation and specified precision;
4. uniqueness, computation and certification of that minimum.

These assertions are not interchangeable. In particular, $\forall x\, \exists D_x$ does not imply $\exists D\, \forall x$ with the same sparse budget. All results below are mathematical statements under their explicit assumptions; they are not new empirical findings about speech.

2 A general existence theorem

Definition 1 (Admissible finite-code representation)**.** Fix a signal space X and shared reconstruction conventions h. An admissible representation is a pair $r = (\theta, z)$ in a declared set $\mathcal{A}_h$, where $z \in \mathbb{F}^m$ for some finite $m \geq 0$, and θ specifies any permitted sample-dependent choice of transform, basis, decoder parameters or side information. A fixed reconstruction rule G_h maps r to X.

Its coefficient cost is

$$c(r) = \|z\|_0 = \#\{j : z_j \neq 0\} \in \mathbb{N}_0.$$

For a declared nonnegative distortion d and tolerance $\varepsilon \geq 0$, set

$$\mathcal{F}_\varepsilon(x) = \{r \in \mathcal{A}_h : d(x, G_h(r)) \leq \varepsilon\}.$$

Exact reconstruction can instead be required directly as $G_h(r) = x$. Any restrictions on adaptation, coefficient range, precision, code length and decoder access are part of $\mathcal{A}_h$ and h.

Lemma 1 (Attainment of an integer-valued minimum)**.** *Let S be nonempty and $c : S \to \mathbb{N}_0$. There exists $s_* \in S$ such that $c(s_*) \leq c(s)$ for all $s \in S$.*

Proof. The image $c(S)$ is a nonempty subset of $\mathbb{N}_0$, so the well-ordering principle gives a least element k_*. Since $k_* \in c(S)$, some $s_* \in S$ satisfies $c(s_*) = k_*$. This s_* has the required property. □

Theorem 2 (Existence of a sparsest feasible finite representation)**.** *For every $x \in X$ and $\varepsilon \geq 0$, if $\mathcal{F}_\varepsilon(x) \neq \varnothing$, then there exists $r_* = (\theta_*, z_*) \in \mathcal{F}_\varepsilon(x)$ satisfying*

$$k_*(x; \varepsilon, \mathcal{A}_h) = \|z_*\|_0 = \min_{(\theta,z)\in\mathcal{F}_\varepsilon(x)} \|z\|_0 .$$

Consequently, any admissible feasible family for every signal in a class $\mathcal{X} \subseteq X$ has a sparsest finite representation for every $x \in \mathcal{X}$.

Proof. Apply Lemma 1 to $S = \mathcal{F}_\varepsilon(x)$ and $c(\theta, z) = \|z\|_0$. □

Remark (What is, and is not, assumed). This result permits infinitely many transforms, noncompact parameter sets and different finite code lengths. It needs no continuity, convexity or compactness: the *attained coefficient counts* are integers. It does require an actual feasible finite code. It establishes neither uniqueness nor an algorithm for finding the minimizer. The minimum distortion for a *fixed* code budget is a different optimization problem and can fail to be attained.

Example 1 (An infimum of error is not feasibility)**.** For scalar target $x = 0$ and fixed decoder $G(z) = e^z$, $z \in \mathbb{R}$, $\inf_z |x - G(z)| = 0$, but $G(z) \neq 0$ for every finite z. At tolerance zero there is no feasible code. At every positive tolerance there is a feasible finite code and Theorem 2 applies. Thus arbitrarily accurate approximations do not, by themselves, prove exact feasibility.

For finite $x \in \mathbb{F}^N$, allowing the identity decoder with $z = x$ proves feasibility with at most N nonzeros. If that decoder is not admissible, feasibility must be established separately. This is the first assumption to check for a learned architecture with an information bottleneck.

3 Unrestricted sequence-dependent bases

3.1 An algebraic statement for every sequence

Lemma 3 (Extension of a linearly independent set to a basis)**.** *Assume the axiom of choice in its equivalent Zorn-lemma form. If V is a vector space over $\mathbb{F}$ and $L \subseteq V$ is linearly independent, then L is contained in an algebraic (Hamel) basis of V.*

Proof. Let $\mathcal{P}$ be the collection of linearly independent subsets of V containing L, ordered by inclusion. It is nonempty since $L \in \mathcal{P}$. For a nonempty chain in $\mathcal{P}$, its union contains L and is linearly independent: a finite linear relation in that union uses finitely many vectors, all contained in one member of the chain. The empty chain has upper bound L. Thus every chain has an upper bound, and Zorn's lemma gives a maximal element B.

If $\operatorname{span}(B) \neq V$, choose $v \notin \operatorname{span}(B)$. Then $B \cup \{v\}$ is linearly independent, contradicting maximality. Hence B spans V and is a basis. □

Theorem 4 (Exact global one-sparsity under unrestricted basis choice)**.** *Let V be a nonzero vector space over $\mathbb{F}$, including the space of length-N finite sequences $\mathbb{F}^N$ or all infinite sequences $\mathbb{F}^{\mathbb{N}}$. Allow every algebraic basis of V, with the basis known to the inverse. Then*

$$\min_{\substack{B \text{ basis of } V \\ x=\sum_{b\in B} z_b b}} \#\{b : z_b \neq 0\} = \begin{cases} 0, & x = 0, \\ 1, & x \neq 0, \end{cases}$$

where the sum is finite, as required for an algebraic basis.

Proof. For $x \neq 0$, the singleton $\{x\}$ is linearly independent. Lemma 3 supplies a basis B_x containing x. Assign coefficient 1 to that basis vector and zero to every other vector; the reconstruction is exact and the cost is one. No linear synthesis of an all-zero coefficient vector can equal a nonzero x, so zero nonzeros are impossible. For $x = 0$, all-zero coefficients attain the lower bound zero. □

Remark (Meaning of the infinite-sequence assertion). For all of $\mathbb{F}^{\mathbb{N}}$, this is an algebraic, generally nonconstructive statement. A Hamel basis need not be countable, computable, continuous or stably encodable. It is not a conventional Fourier/wavelet expansion or a finite-bit codec. The theorem explicitly uses a basis that may depend on the target sequence.

3.2 Finite sequences: a constructive orthonormal result

Lemma 5 (Finite orthonormal completion)**.** *For any unit vector $u \in \mathbb{F}^N$, there exists an orthonormal basis $(u, q_2, \ldots, q_N)$ of $\mathbb{F}^N$.*

Proof. Run Gram–Schmidt on the ordered spanning list $(u, e_1, \ldots, e_N)$, discarding zero residuals. Its first output is u. The process preserves the span of processed vectors and produces orthonormal nonzero outputs. Since the original list spans $\mathbb{F}^N$, exactly N outputs are obtained, giving the required basis. This finite construction needs no infinite-dimensional basis-extension argument. □

Corollary 6 (Finite orthonormal sparsest-domain existence)**.** *For every nonzero $x \in \mathbb{F}^N$, there is a unitary synthesis matrix Q_x such that*

$$Q_x^* x = \|x\|_2 \, e_1, \qquad x = Q_x(\|x\|_2 \, e_1).$$

The minimum nonzero count over all invertible linear coordinate systems, and also over the smaller family of unitary coordinate systems, is exactly one. For $x = 0$, both minima are zero.

Proof. Apply Lemma 5 to $u = x/\|x\|_2$ and let Q_x have the resulting orthonormal vectors as columns. Inner products with these columns give the displayed coefficient vector. The zero-code lower bound follows from linearity, as in Theorem 4. □

Corollary 7 (Unrestricted best sparsity at a norm-error tolerance)**.** *For $x \in \mathbb{F}^N$ and distortion $\|x - Qz\|_2$, if all unitary Q are allowed,*

$$k_*(x; \varepsilon) = \begin{cases} 0, & \|x\|_2 \leq \varepsilon, \\ 1, & \|x\|_2 > \varepsilon. \end{cases}$$

Proof. Zero coefficients reconstruct zero with error $\|x\|_2$. If that error is inadmissible, at least one coefficient is necessary, and Corollary 6 gives an exact one-coefficient reconstruction. □

Thus even orthogonality, norm preservation and condition number one do not stop the basis from encoding the entire direction of x. These are not proposed constraints for our model; they show that the issue persists under strong linear regularity.

3.3 Finite-energy infinite sequences

Theorem 8 (One-sparsity in a sequence-dependent Hilbert basis)**.** *For every nonzero $x \in \ell^2(\mathbb{N};\mathbb{F})$ there is a countable complete orthonormal basis $(q_j)_{j\geq 1}$ of ℓ^2 such that*

$$q_1 = x/\left\|x\right\|_2, \qquad x = \left\|x\right\|_2 q_1.$$

Hence the exact minimum over these sequence-dependent Hilbert bases is one.

Proof. Apply Gram–Schmidt to $(x/\left\|x\right\|_2, e_1, e_2, \ldots)$, skipping zero residuals. The resulting orthonormal sequence has the same closed linear span as this list. The list contains the canonical basis of ℓ^2, whose finite linear span is dense, so the resulting sequence is complete. Its first element is $x/\left\|x\right\|_2$, giving the claimed expansion. The zero-code lower bound is again linearity. □

Proposition 9 (Affine adaptation makes the minimum zero)**.** *If invertible affine coordinate systems may be chosen separately for each $x \in \mathbb{F}^N$, every x has an exact representation with zero nonzeros.*

Proof. Choose $T_x(y) = y - x$, with inverse $T_x^{-1}(z) = z + x$. Then $T_x(x) = 0$ and $T_x^{-1}(0) = x$. Nonzero count is nonnegative, so zero is optimal. □

Interpretation. The linear theorem stores the sequence in its basis; the affine theorem stores it in an offset. The inverse must know this sample-dependent information. Neither proves economical reconstruction from the counted coefficient vector alone.

4 A nontrivial admissible family: a fixed finite dictionary

Here the dictionary is fixed before the target is represented. These statements do not optimize over every possible dictionary.

Lemma 10 (Best approximation in a finite-dimensional subspace)**.** *Let $W \subseteq \mathbb{F}^N$ be a linear subspace. There is a unique $p_W x \in W$ such that*

$$\left\|x - p_W x\right\|_2 = \min_{y\in W} \left\|x - y\right\|_2.$$

Proof. Choose an orthonormal basis $(u_1, \ldots, u_r)$ of W and put $p_W x = \sum_{j=1}^{r} \langle x, u_j\rangle u_j$, using the inner-product convention linear in its first argument. The residual is orthogonal to W. For any $y \in W$,

$$\left\|x - y\right\|_2^2 = \left\|x - p_W x\right\|_2^2 + \left\|p_W x - y\right\|_2^2.$$

This proves both minimality and uniqueness. For $W = \{0\}$, take $p_W x = 0$. □

Theorem 11 (Existence and exact characterization in a fixed dictionary)**.** *Fix $D \in \mathbb{F}^{N\times M}$, $M \geq 1$. For $S \subseteq \{1, \ldots, M\}$ write $W_S = \operatorname{span}\{D_j : j \in S\}$, with $W_\varnothing = \{0\}$. Define*

$$\rho_k(x) = \min_{\substack{S\subseteq\{1,\ldots,M\}\\ |S|\leq k}} \operatorname{dist}(x, W_S), \qquad 0 \leq k \leq M.$$

Each minimum is attained and equals

$$\rho_k(x) = \min_{\|z\|_0 \le k} \|x - Dz\|_2 .$$

A feasible code at tolerance ε exists exactly when $\operatorname{dist}(x, \operatorname{range} D) \le \varepsilon$. *When feasible, a sparsest code exists and its cost is*

$$k_* = \min\{k : \rho_k(x) \le \varepsilon\}.$$

In particular, if D spans $\mathbb{F}^N$, every finite sequence has a sparsest exact representation in D, with at most N nonzeros.

Proof. For each support S, Lemma 10 gives an attained best waveform $p_{W_S}x$. Since it belongs to the column span of D_S, it has coefficients supported on S, even if these coefficients are nonunique. There are finitely many supports with $|S| \le k$, so a support attaining the smallest error exists. Conversely, every k-sparse code synthesizes into one of these subspaces. This proves the equality and attainment of ρ_k.

Taking all columns gives the feasibility criterion. Feasibility makes the displayed set of integers nonempty, and its least element is attained by the preceding construction. If D spans $\mathbb{F}^N$, choose N independent columns forming a basis; they represent every x exactly. □

No coefficient bound or compact parameter space is needed here: each support produces a closed finite-dimensional waveform subspace, so its best waveform approximation is attained. Exhaustive support search can be expensive; existence is not a claim of efficient computation.

Corollary 12 (Certified best basis in a finite orthonormal library)**.** *Let $(Q_\ell)_{\ell=1}^J$ be a fixed finite library of unitary $N \times N$ matrices. For each ℓ, let $a^{(\ell)} = Q_\ell^* x$, and order its magnitudes as $|a^{(\ell)}|_{(1)} \ge \cdots \ge |a^{(\ell)}|_{(N)}$. Then*

$$\min_{\ell, \|z\|_0 \le k} \|x - Q_\ell z\|_2^2 = \min_{1 \le \ell \le J} \sum_{j=k+1}^{N} |a^{(\ell)}|_{(j)}^2 .$$

The least k for which the right side is at most ε^2 is the globally sparsest feasible code in this declared library.

Proof. For fixed ℓ, unitarity gives $\|x - Q_\ell z\|_2^2 = \left\|a^{(\ell)} - z\right\|_2^2$. On a chosen support the best coefficients equal the corresponding entries of $a^{(\ell)}$. Keeping the k largest squared magnitudes minimizes the discarded sum. A finite comparison over ℓ attains the best result. □

This supplies a rigorous small oracle for later experiments. It is a certificate within the library, not across unknown transforms outside it. The library identifier is part of the reconstruction information when the chosen basis is not otherwise known to the inverse.

Theorem 13 (Conditional uniqueness via spark)**.** *For a fixed finite dictionary D, let* $\operatorname{spark}(D)$ *be the smallest number of linearly dependent columns, with convention* $\operatorname{spark}(D) = M + 1$ *if all columns are independent. If $Dz = x$ and $2\|z\|_0 < \operatorname{spark}(D)$, then z is the unique sparsest exact code in D.*

Proof. Put $k = \|z\|_0$. If another code $w \ne z$ satisfies $Dw = x$ and $\|w\|_0 \le k$, then $v = z - w \ne 0$ satisfies $Dv = 0$ and has at most $2k$ nonzeros. Its nonzero coordinates give a linear dependence among fewer than $\operatorname{spark}(D)$ columns, a contradiction. Thus no distinct code of cost at most k exists. □

This standard result is included with its proof to separate *existence* from *uniqueness*; see Donoho and Elad [1]. It is not a noisy recovery theorem, a necessary uniqueness condition, or a certificate that D is the best dictionary over all admissible choices.

5 Why a shared stable decoder is a different claim

Lemma 14 (Lipschitz images of lower-dimensional domains have zero volume)**.** *If $f : \mathbb{R}^s \to \mathbb{R}^N$ is Lipschitz and $0 \le s < N$, then $f(\mathbb{R}^s)$ has zero N-dimensional Lebesgue measure.*

Proof. For $s = 0$, the image is a point and has zero volume. For $s \ge 1$, restrict to $[-R, R]^s$. Partition it into n^s cubes of side $2R/n$. If $L > 0$ is a Lipschitz constant, the image of each cube has diameter at most $2LR\sqrt{s}/n$, and fits in an N-dimensional box of side at most $4LR\sqrt{s}/n$. The total volume of these covering boxes is bounded by

$$n^s(4LR\sqrt{s}/n)^N = (4LR\sqrt{s})^N n^{s-N} \longrightarrow 0.$$

The image therefore has zero outer measure. The case $L = 0$ is constant. Finally $\mathbb{R}^s = \bigcup_{R\in\mathbb{N}}[-R, R]^s$, and a countable union of null sets is null. □

Theorem 15 (No universal exact low-support code under a shared Lipschitz decoder)**.** *Fix $G : \mathbb{R}^M \to \mathbb{R}^N$ Lipschitz and an integer $0 \le k < N$. Then*

$$\mathcal{Y}_k = \{G(z) : \|z\|_0 \le k\}$$

has zero N-dimensional measure and contains no nonempty open subset of $\mathbb{R}^N$. Consequently, one such decoder cannot exactly reconstruct every waveform in an open full-dimensional class using at most k nonzeros.

Proof. For each support $S \subseteq \{1, \dots, M\}$ with $|S| \le k$, the coordinate embedding $\iota_S : \mathbb{R}^{|S|} \to \mathbb{R}^M$ is an isometry. Thus $G \circ \iota_S$ is Lipschitz and its image is null by Lemma 14. There are finitely many supports. Their union is $\mathcal{Y}_k$, which is null. Every nonempty open subset of $\mathbb{R}^N$ contains a box with positive volume, so none lies in $\mathcal{Y}_k$. □

A finite collection of such decoders also has a null union of k-sparse outputs. If a single decoder receives p additional continuous adaptation parameters and is jointly Lipschitz, the same proof applies when $p + k < N$, using $\mathbb{R}^p \times \mathbb{R}^{|S|}$ as the parameter domain. Adaptation cannot be ignored in a dimension count.

This result does not contradict Theorem 4, whose basis may depend freely on x. It also does not assert that speech contains an open subset of waveform space. Nor does null volume by itself provide a positive *approximation* error bound: a null set may be dense. Approximate, stable compression needs further code-range, precision and model-class assumptions. Class-level stable approximation is treated more generally by Cohen et al. [2]; their work is not a per-sequence shortest-code certificate.

6 Infinite sequences and finite-precision reconstruction

Example 2 (No finite exact code in a fixed infinite coordinate basis)**.** Let $x = (2^{-n})_{n\ge1} \in \ell^2$ and use its fixed canonical basis (e_n). Every coordinate of its unique exact expansion is nonzero, so no finite-support exact representation exists in that basis. Theorem 2 does not supply one because its exact finite-code feasible set is empty.

For support S, the best coefficient values are the original entries; the captured energy is $\sum_{n\in S} 4^{-n}$, maximized among k-element supports by the first k indices. Thus for any positive norm-error tolerance ε, the best k-term canonical approximation has squared error

$$\sum_{n=k+1}^{\infty} 4^{-n} = \frac{4^{-k}}{3}.$$

Hence

$$k_* = \max\left(0, \left\lceil \log_4 \frac{1}{3\varepsilon^2} \right\rceil\right)$$

is finite and attained. At $\varepsilon = 0$ no finite k suffices. By Theorem 8, the very same sequence is one-sparse in a suitably chosen sequence-dependent Hilbert basis. The admissible family changes the answer.

Theorem 16 (A finite-alphabet counting bound)**.** *Consider all length-N sequences over an alphabet of size $q \geq 2$. Suppose a fixed reconstruction system accepts a choice from J declared models and a code z of M coordinates, each from an alphabet of $L \geq 2$ coefficient symbols including zero, with at most $k \leq M$ nonzeros. If it reconstructs all q^N sequences exactly, it is necessary that*

$$q^N \;\leq\; J \sum_{j=0}^{k} \binom{M}{j} (L-1)^j.$$

Proof. There are $\binom{M}{j}$ choices of a support of size j and $(L-1)^j$ assignments of nonzero symbols on it. Multiplying the sum by J bounds the total number of possible input descriptions to the fixed decoder. A deterministic decoder produces at most that many distinct outputs. To reconstruct q^N distinct sequences exactly, it must have at least q^N distinct possible outputs. □

The support index and model choice are included in this bound. Increasing coefficient precision or allowing uncounted sequence-specific models can evade a coefficient-count comparison without reducing the actual information supplied. This motivates explicit model/precision conventions, consistent with the description-accounting perspective of Rissanen [3]; it does not replace minimum support with a different objective.

Proposition 17 (No finite-bit exact description for every infinite binary sequence)**.** *Under any fixed decoder whose only input is a finite binary string, not every infinite binary sequence has an exact description.*

Proof. Finite binary strings form a countable set, so the decoder has at most countably many outputs. Infinite binary sequences are uncountable: if $(s^{(j)})_{j\geq 1}$ were an enumeration, the sequence $t_j = 1 - s_j^{(j)}$ would differ from its jth listed sequence at coordinate j, contradicting completeness of the list. Thus some infinite binary sequences are absent from the decoder's range. □

An algebraic one-coefficient representation from Theorem 4 does not contradict this proposition: its arbitrary basis and exact scalar need not have finite binary descriptions. Likewise, mathematical finite real-valued vectors and finite-precision stored recordings are distinct objects.

7 What the existence proof establishes for this project

The strongest unrestricted conclusion is already exact:

> Every nonzero sequence admits an exact one-sparse representation in a sequence-dependent algebraic basis. Every feasible finite-code problem has an attained sparsest code within its declared admissible family. Neither statement identifies a useful universal domain or an efficiently discoverable optimum.

For finite audio, our meaningful research target remains

$$\min_{\theta,z} \|z\|_0 \quad \text{subject to} \quad d(x, G_h(\theta, z)) \leq \varepsilon, \qquad (\theta, z) \in \mathcal{A}_h.$$

The family may include unknown learned or sample-adapted domains. Its rules must declare what the inverse knows, the allowed adaptation information, coefficient precision/range, and any stability requirements. Existence does not force a particular dictionary, orthogonality, semantic interpretation, or model architecture.

A feasible k-coefficient code gives an upper bound $U = k$ on the optimum. A lower bound L must rule out *all* allowed codes and transform choices with fewer coefficients. Matching bounds $L = U$ certify optimality within the family. A unique sparsest code for one fixed dictionary does not prove that dictionary is the best possible domain.

The frozen S-AF autoencoder is one admissible encoder–decoder construction. Its successful reconstructions exhibit achievable codes. They do not prove global sparsest-domain optimality. The next theoretical decision is the admissible family and information-accounting rules; training can then be studied as a search mechanism for that defined objective.

References

[1] D. L. Donoho and M. Elad. *Optimally sparse representation in general (nonorthogonal) dictionaries via ℓ^1 minimization*. Proceedings of the National Academy of Sciences, 100(5):2197–2202, 2003. doi:10.1073/pnas.0437847100.

[2] A. Cohen, R. DeVore, G. Petrova and P. Wojtaszczyk. *Optimal Stable Nonlinear Approximation*. arXiv:2009.09907, 2020. https://arxiv.org/abs/2009.09907.

[3] J. Rissanen. *Modeling by shortest data description*. Automatica, 14(5):465–471, 1978. doi:10.1016/0005-1098(78)90005-5.